

Chapter 32

Optimal Unit Locations in Emergency Service Systems with Bayesian Optimization



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Abstract We model the facility location problem in an emergency service system as an optimization problem in which the objective is to minimize the system-wide mean response time, which requires exponential complexity to solve. We show that this problem is NP-hard and develop lower and upper bounds for the optimal solution from a special case of the classical p-median problem. We propose a Bayesian optimization solution to this problem that includes searching within feasible trust regions and adaptive swapping strategies. We show that our algorithm always converges to a globally optimal solution with a regret bound guarantee. Our algorithm consistently outperforms the p-median solution in numerical experiments and quickly converges to the optimal solution. We also apply our method to solve the optimal ambulance location problem in St. Paul, Minnesota, using one year of real data. We show that our method converges to the optimal solution very quickly. Our method can be applied to solve the optimal unit locations in the emergency service systems of the largest cities.

Keywords Facility location · Bayesian optimization · Combinatorial optimization · Emergency service system

Introduction

In most emergency service systems, system-wide mean response time is a crucial system performance indicator. The response time for a call in an emergency system begins when the system receives this call and ends when the responder arrives at the service location, defined as the sum of the turnout and travel times. By reducing response times, both patients and healthcare providers can benefit. For the patient,

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shorter response times improve the chances of survival in an emergency and the quality of the subsequent medical care. For service providers, shorter response times save service time and make the system more efficient. However, as we will show in the paper, solving the exact mean response time given a specific configuration requires solving a linear system whose size grows exponentially with the number of units, which is proven to be NP-hard, let alone solving for the best configuration. Therefore, this problem is usually studied in a p-median setting by ignoring the state transitions and assuming all units are always free whenever calls are received in the system, which greatly simplifies the problem.

This study models the facility location problem in an emergency service system to minimize the average response time as a combinatorial optimization problem that is NP-hard. We show that the value of the optimal solution in a particular p-median problem [5] is a lower bound of our problem, and we also obtain an upper bound from this particular p-median problem. We propose a Bayesian optimization scheme for this problem, which includes using feasible trust regions and adaptive swapping to solve for the optimal solution. We show that our algorithm always converges to the global optimum with a regret bound guarantee. We show that our algorithm always beats the p-median solution and converges to the optimal solution very quickly in numerical experiments. We also apply our method to solve the optimal unit location in St. Paul, Minnesota, using one year of real data. We found our method converges to the optimal solution quickly.

Literature Review

Facility location problems in healthcare are widely studied. Ahmadi-Javid et al. [1] provides a comprehensive review of facility location problems in healthcare systems. Li et al. [9] reviews the literature on coverage models and optimization techniques for emergency facility locations. Daskin [3] provides a comprehensive review of discrete location problems and classifies them into three main categories: coverage-based problems, median-based problems, and other problems.

The p-median location problem is one of the most famous problems in facility location. The p-median problem was first proposed by Hakimi [5]. It was shown to be NP-hard even with a simple structure by Kariv and Hakimi [6]. Much of the work in the literature models the location problem of emergency medical facilities as a p-median problem. Rajagopalan and Saydam [10], minimizes the expected response distance of the entire system while satisfying the coverage requirement, by a heuristic tabu search approach. However, their solution has no performance guarantee and is slow to converge.

The p-median problem simplifies the computation of response times, while the exact average response time can be solved by the spatial hypercube model, that was developed by Larson [7]. The hypercube model is used to analyze a system consisting of spatially distributed service units, such as ambulances or fire trucks. This hypercube model and its extensions have been showed to be significant in evaluating

performance of emergency service systems. For instance, it has been implemented in the maximal expected covering location problem Batta et al. [2]. This model entails solving 2^p simultaneous equations for an p -server system, where each server is either busy or available. Larson [8] developed an approximation for larger systems, that reduces the number of equations to p . We will use this approximation algorithm as an approximation to the response time.

Bayesian optimization is a method for optimizing an objective function that requires a relatively long time to evaluate. It also tolerates random noise in function evaluation, such as approximation algorithms or simulations [4]. It creates a surrogate function for the objective, uses Bayesian techniques to quantify the uncertainty of that surrogate function, and then uses the acquisition function defined from that surrogate function to determine the next point to evaluate. The surrogate function is then updated upon evaluation of the evaluated point to form a better belief of the actual function.

Model

This section develops a model for solving the minimum average response time for unit locations in emergency service systems. We show that this problem is NP-hard and develop lower and upper bounds for this problem.

Problem Definition

Our goal in this study is to select the best p unit locations in an emergency service system with N candidate locations to minimize the average response time of all calls. We let I be the set of all candidate locations and $N = |I|$ be the cardinality of I , the total number of candidate locations. The entire region is divided into $|J|$ subregions, where J is the set of all subregions. Calls are generated from each subregion, and the call arrival rate for subregion j is assumed to follow a Poisson distribution with rate λ_j . The region can be divided as finely as desired without increasing the complexity of the problem. We let μ be the average service rate of all units. Calls are assigned to each unit according to a fixed assignment rule (which can be any general assignment policy) that considers the availability of each unit. If all units are busy when the call arrives, we assume it is handled by a unit outside the system, either a private emergency service unit or a unit from an adjacent jurisdiction, which is a common mutual aid policy in emergency service systems. As mentioned in the Introduction, the response time is the sum of the unit turnout time from the base location and the travel time to the scene. We let t_{ij} be the average travel time from location i to call location j and let τ_o be the average turnout time.

Let x_i be the binary variable indicating whether location i is selected or not, where $x_i = 1$ means location i is occupied by a unit, and $x_i = 0$ indicates location i is empty. We let $\mathbf{x} = \{x_1, \dots, x_N\}$ be the location decision set. Our goal is to solve for the optimal location decision set $\mathbf{x}^*$ that minimizes the mean response time of the system. The optimal set $\mathbf{x}^*$ can be obtained by solving the following integer programming, which we denote as *p-MRT*.

$$\mathbf{x}^* = \arg \min_{\mathbf{x}} \sum_{i \in I} \sum_{j \in J} t_{ij} q_{ij}(\mathbf{x}) \quad (\text{p-MRT}) \quad (1)$$

$$\text{s.t. } \sum_{i \in I} x_i = p, \quad (2)$$

$$q_{ij}(\mathbf{x}) = \frac{\sum_{\mathbf{b} \in S_{ij}(\mathbf{x})} \lambda_j P(\mathbf{b})}{\sum_{j \in J} \lambda_j (1 - \Pi)}, \quad \forall i \in I, \forall j \in J, \quad (3)$$

$$x_i \in \{0, 1\}, \quad \forall i \in I, \quad (4)$$

where $q_{ij}(\mathbf{x})$ is the fraction of all calls in subregion j that are responded to by the unit at location i given unit locations $\mathbf{x}$. The objective is to minimize the system-wide mean response time. The first constraint indicates that there are a total of p units to be located, and the second constraint shows the expression for obtaining $q_{ij}(\mathbf{x})$ from the spatial hypercube model. The state $\mathbf{b} = \{b_1, \dots, b_p\}$ denotes the state of all p units, where $b_k = 0$ means that unit k is free and $b_k = 1$ means that unit k is busy with a service call. $P(\mathbf{b})$ is the probability of being in state $\mathbf{b}$, and Π is the probability that all units are busy, also known as the blocking probability. $S_{ij}(\mathbf{x})$ is the set of states in which the unit at location i is assigned to calls from subregion j , which is fixed under the assignment policy. Note that the exact values of $q_{ij}(\mathbf{x})$ requires knowing $P(\mathbf{b})$ and Π , which require solving the following balance equation of the spatial hypercube model. We have, for all $\mathbf{b} \in \{0, 1\}^p$,

$$P(\mathbf{b}) \left[\sum_{j \in J} \lambda_j + \sum_{k=1}^p b_k \mu \right] = \sum_{\mathbf{b}': d(\mathbf{b}, \mathbf{b}')=1} P(\mathbf{b}') r(\mathbf{b}, \mathbf{b}') + \sum_{\mathbf{b}': d(\mathbf{b}', \mathbf{b})=1} P(\mathbf{b}') \mu, \quad (5)$$

where $r(\mathbf{b}, \mathbf{b}')$ is the transition rate from state $\mathbf{b}$ to state $\mathbf{b}'$ in the spatial hypercube model, and $d(\mathbf{b}, \mathbf{b}')$ is the Hamming distance from $\mathbf{b}$ to $\mathbf{b}'$. The spatial hypercube model was originally developed by Larson [7], which contains 2^p simultaneous equations as described in Eq. (5).

The following theorem shows that the *p-MRT* problem is NP-hard, and the proof is in the full version of the paper.

Theorem 1 *The p-MRT problem is NP-hard.*

Performance Bounds

In this section, we develop lower and upper bounds for the optimal solution of the p -MRT problem by relating it to the classical p -median problem. The p -median problem, proposed by Hakimi [5], solves the minimum total demand-weighted response time in a simplified setting that ignores the stochasticity of the system.

We first develop the lower bound. In the p -median problem, $x_j \in \{0, 1\}$ is a binary variable indicating the unit location as in the p -MRT problem, and $y_{ij} \in \{0, 1\}$ is a binary variable indicating whether calls from subregion j are assigned to the unit at location i . We let $(\tilde{x}^*, \tilde{y}^*)$ denote the optimal solution to the p -median problem. The goal of the p -median problem is to minimize the total demand-weighted response time.

$$(\tilde{x}^*, \tilde{y}^*) = \arg \min_{x, y} \sum_{i \in I} \sum_{j \in J} w_j t_{ij} y_{ij} \quad (\text{p-median}) \quad (6)$$

$$\text{s.t. } \sum_{i \in I} y_{ij} = 1, \forall j \in J, \quad (7)$$

$$\sum_{i \in I} x_i = p, \quad (8)$$

$$y_{ij} \leq x_i, \forall i \in I, \forall j \in J, \quad (9)$$

$$y_{ij} \in \{0, 1\}, \forall i \in I, \forall j \in J, \quad (10)$$

$$x_i \in \{0, 1\}, \forall i \in I, \quad (11)$$

The first constraint indicates that each subregion will be assigned to a unit, and the third constraint indicates that calls in subregion j can only be assigned to a location with a unit. Note that the p -median problem does not include the service rate μ because its underlying assumption ignores state transitions and assumes that all units are always available, which is equivalent to assuming that each unit has a service rate of $\mu \rightarrow \infty$, and all units become immediately available for the next call after being assigned to one call.

In the following theorem, we show that the optimal value of the p -MRT problem is lower bounded by the optimal value of a special case of the p -median problem. The detailed proof of the theorem is in the full paper.

Theorem 2 When setting $w_j = \lambda_j / \sum_{j \in J} \lambda_j$, the optimal value to the p -median problem is a lower bound on the optimal value of the p -MRT problem.

Next, we develop a feasible upper bound from the same p -median problem as in Theorem 2. It can be shown that the optimal unit locations $\tilde{x}^*$ from the above p -median problem is always a feasible solution to the p -MRT problem, because it satisfies the constraint $\sum_{i \in I} x_i = p$ of the p -MRT problem. Since the p -MRT problem is a minimization problem, every feasible solution provides an upper bound

on its optimal value. Therefore, when setting $w_j = \lambda_j / \sum_{j \in J} \lambda_j$, the optimal location $\tilde{\mathbf{x}}^*$, obtained from the p -median problem, provides an upper bound on the optimal value to the p -MRT problem.

Bayesian Optimization Solution Approach

Bayesian optimization is an optimization method for objective functions that require a relatively long time to evaluate or lack special structure. It also tolerates random noise in function evaluation, such as complex objective functions that can be evaluated only using approximation algorithms or simulations. It generally consists of a surrogate model and an acquisition function based on the surrogate model. This section presents a Bayesian optimization solution for the p -MRT problem. We also show the convergence and regret bounds of our method.

Surrogate Model

We let $f(\mathbf{x}) = \sum_{i \in I} \sum_{j \in J} t_{ij} q_{ij}(\mathbf{x})$ be the exact mean response time obtained from the spatial hypercube model, as shown in Section “[Model](#)”, and solving it requires extensive computation. To handle this problem, we can use simulation or approximation algorithms to obtain an estimate of $f(\mathbf{x})$, and we will discuss the approximation algorithm in detail in Section “[Spatial Hypercube Approximation](#)”. We let $y = f(\mathbf{x}) + \epsilon$ be the estimated mean response time, where $\epsilon \sim \mathcal{N}(0, \sigma^2)$ is the estimation error and is assumed to follow a Gaussian distribution.

We use the Gaussian process as the surrogate model. We use $\mathbf{X}$ to denote the set containing of all evaluated points (each point represents the locations of a set of p units), and $\mathbf{y}$ is the set of the corresponding objective values (realized mean response time) of these points. A Gaussian process $GP(m(\mathbf{X}), k(\mathbf{X}, \mathbf{X}'))$ is specified by its mean function $m(\mathbf{X}) = E[f(\mathbf{X})]$ and the covariance (or kernel) function $k(\mathbf{X}, \mathbf{X}') = E[(f(\mathbf{X}) - m(\mathbf{X}))(f(\mathbf{X}') - m(\mathbf{X}'))]$. The distribution of the Gaussian process is the joint distribution of all currently evaluated points, and each finite subset of all these points follows a multivariate normal distribution. Given the existing set of points $\mathbf{X}$, for any point of interest $\tilde{\mathbf{x}}$, its posterior distribution follows the normal distribution $\mathcal{N}(\mu(\tilde{\mathbf{x}}), \sigma^2(\tilde{\mathbf{x}}))$ and $\mu(\tilde{\mathbf{x}})$ and $\sigma^2(\tilde{\mathbf{x}})$ are the mean and variance, where

$$\mu(\tilde{\mathbf{x}}) = m(\tilde{\mathbf{x}}) + k(\tilde{\mathbf{x}}, \mathbf{X})[k(\mathbf{X}, \mathbf{X}) + \sigma_n^2 I]^{-1}(\mathbf{y} - m(\mathbf{X})), \quad (12)$$

$$\sigma^2(\tilde{\mathbf{x}}) = k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) - k(\tilde{\mathbf{x}}, \mathbf{X})[k(\mathbf{X}, \mathbf{X}) + \sigma^2 I]^{-1}k(\mathbf{X}, \tilde{\mathbf{x}}). \quad (13)$$

We use the transformed overlap kernel $k(\mathbf{x}, \mathbf{x}') = \exp\left(\sum_{i=1}^N \ell_i \delta(x_i, x'_i) / N\right)$, introduced by Wan et al. [12], where ℓ_i is the lengthscale of the kernel.

Acquisition Function

Due to the high cost of evaluating a unit location set, we want to find the best location set with fewer evaluations. Therefore, in Bayesian optimization, data are sequentially acquired and added to the Gaussian Process (GP) model, and the acquisition function is designed to strategically select the next location set to be evaluated. In contrast, obtaining the optimal solution by complete enumeration does not use any known information about the evaluated points.

In our solution framework, we search within the feasible trust region, as introduced in Section “Feasible Trust Region”, and move to a new feasible trust region using a restart strategy with a logic similar to the restart algorithm in the non-stationary bandit problem. An acquisition function chooses the center of the new feasible trust region based on the GP model. After each restart, we fit the GP model on (X_t, Y_t) , where t denotes the number of restarts and each $\mathbf{x} \in X_t$ is the local optimal solution obtained in the feasible trust region in each restart round. We use a Lower Confidence Bound (LCB) acquisition function to select the new center of the trust region at the beginning of each restart.

$$\mathbf{x}_{t+1} = \underset{\mathbf{x}}{\operatorname{argmin}} \mu(\mathbf{x}; X_t, Y_t) - \beta_t^{1/2} \sigma(\mathbf{x}; X_t, Y_t), \quad (14)$$

where $\mu(\mathbf{x}; X_{t-1}, Y_{t-1})$ and $\sigma(\mathbf{x}; X_{t-1}, Y_{t-1})$ are the mean and standard deviation of the GP learned from the observed data (X_{t-1}, Y_{t-1}) .

Feasible Trust Region

Our algorithm defines the feasible trust region by the Hamming distance as shown in (15).

$$\operatorname{TR}_d(\mathbf{x}^c) = \left\{ \mathbf{x} \mid \sum_{i=1}^N \delta(x_i, x_i^c) \leq d \text{ and } \sum_{i=1}^N x_i = p \right\}, \quad (15)$$

where δ is the Kronecker delta function.

Updating within the feasible trust region ensures that the next candidate solution always satisfies the constraints and is within the trust region. We apply the adaptive swapping search strategy within the feasible region to find the next candidate solution. The adaptive swapping search strategy performs $s(d)$ rounds of random swapping of an empty location and a location with a unit on the point $\mathbf{x}$ with the current best acquisition function with in the trust region, where $s(d)$ is a function of the current length of the trust region d .

Given a trust region center $\mathbf{x}^c = \{x_1, \dots, x_N\}$, we swap the value of each binary variable pair $\{x_i, x_j\} \in \mathbf{x}^c$ if one of them has value 1 and the other has value 0. By doing so, we ensure that the constraint of the p-MRT problem remains satisfied.

Consider the trust region length to be d , then we can conduct this swapping operation up to $\lfloor d/2 \rfloor$ times to ensure our searching space lies within the trust region. The trust region length d shrinks by a factor of $\alpha_s \geq 1$ if the adaptive swapping search algorithm fails to improve the value of the acquisition function with a consecutive of n_f trials and expands by a factor $\alpha_e \geq 1$ if the total number of successes reaches n_s .

As mentioned earlier, the restart mechanism is designed to prevent the algorithm from getting stuck in a local optimum, as in Wan et al. [12]. In our algorithm, we initiate a restart whenever $\lfloor d \rfloor < 2$. We evaluate the local optimum obtained based on the objective function, add it to the GP model and update the model. The trust region center for the next restart is obtained from the updated GP model.

Theoretical Results

We denote $r_t = f(\mathbf{x}_t) - f(\mathbf{x}^*)$ as the regret for the t -th evaluation of the objective function, with the total regret up to time T as $R_T = \sum_{t=1}^T r_t$.

Theorem 3 *Our algorithm converges to the global optimum of p -MRT in a finite number of iterations.*

Theorem 4 *The regret bound of our algorithm is, for any $\zeta \in (0, 1)$,*

$$\Pr \left\{ R_T \leq \sqrt{16NT\beta_T\kappa\gamma_T} \right\} \geq 1 - \zeta, \quad (16)$$

where $\beta_T = \log(2T^2\pi^2/6\zeta)$ and $\kappa = \sigma^{-2}/\log(1 + \sigma^{-2})$.

The detailed proofs of the above two theorems are in the full paper. We give here a brief roadmap of these two proofs. Theorem 3 is proved by contradiction. We show that the size of the feasible trust region d will always be reduced to less than two and that the exploration is complete for the trust region. The proof of Theorem 4 follows a logic similar to that of the GP-UCB algorithm, as shown in Srinivas et al. [11]. The γ_T in the above theorem is the maximum information gain of the acquisition function under the proposed kernel function studied in Wan et al. [12].

Spatial Hypercube Approximation

The Bayesian optimization approach requires evaluating the objective function value of a given location set $\mathbf{x}$, which is $\sum_{i \in I} \sum_{j \in J} t_{ij} q_{ij}(\mathbf{x})$. Obtaining the exact value, however, requires solving the spatial hypercube model developed by Larson [7], which has an exponential complexity. This section briefly describes the approximation algorithm developed by Larson [8] that we use to approximate the mean response time objective function.

Algorithm 1 Bayesian Optimization with Feasible Trust Region

```

1: Input: evaluating function  $f(x)$ ; initial sample size  $I$  and trust region  $TR_0$ ;
   number of iterations  $K$ .
2:  $t \leftarrow 1, X_0 \leftarrow \emptyset, Y_0 \leftarrow \emptyset$ .
3:  $\text{restart} \leftarrow \text{True}$ .
4: while  $t < K$  do
5:   if  $\text{restart}$  then
6:     Randomly select  $I$  initial samples  $X_I = \{x_1, \dots, x_I\}$  within  $TR_{t-1}$ .
7:     Update observed data  $X_t \leftarrow X_{t-1} \cup X_I, Y_t \leftarrow Y_{t-1} \cup f(X_I)$ 
8:     Initialize the surrogate model with current observations  $\hat{f} \leftarrow (X_t, Y_t)$ .
9:      $\text{restart} \leftarrow \text{False}$ .
10:  else
11:    Construct a trust region  $TR_t$  around the best solution found  $x_t^* \in X_t$  using
    Eq. (14).
12:    Select next candidate  $z_t$  within  $TR_t$  by the surrogate model  $z_t =$ 
     $\text{argmin}_{x \in TR(x_t^*)} \hat{f}(x)$ .
13:    Evaluate  $f(z_t)$  and record the observation  $y_t$ .
14:    Update observed data  $X_t \leftarrow X_{t-1} \cup \{z_t\}, Y_t \leftarrow Y_{t-1} \cup \{y_t\}$  and the surro-
    gate model  $\hat{f} \leftarrow (z_t, y_t)$ .
15:    if  $y_t^* = \min(Y_t)$  keeps unchanged for a certain iterations then
16:       $\text{restart} \leftarrow \text{True}$ .
17:       $t \leftarrow t + 1$ .
Output: Best solution found  $x_t^*$ .

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Define p_{ijk} as the probability that unit i is dispatched as the k -th preferred unit at node j . Unit i will be dispatched only if it is available and all $k - 1$ units more preferred are busy. We have

$$p_{ijk} = \Pr \left[W_i^c \cap \left\{ \bigcap_{l=1}^{k-1} W_{\gamma_{jl}} \right\} \right], \quad (17)$$

where W_i is the event that unit i is busy, and W_i^c is the event that unit i is available.

Recall that the rate that a call arrives at node j is λ_j . Then, the rate at which unit i is dispatched to node j as the k -th preferred unit is $\lambda_j p_{ijk}$, and the rate, S_i , at which unit i is dispatched to calls from all demand nodes is

$$S_i = \sum_{j \in G_i^1} \lambda_j p_{ij1} + \sum_{j \in G_i^2} \lambda_j p_{ij2} + \dots + \sum_{j \in G_i^N} \lambda_j p_{ijN} = \sum_{k=1}^N \sum_{j \in G_i^k} \lambda_j p_{ijk}, \quad (18)$$

where G_i^k is defined as the set of nodes for which unit i is the k -th preferred unit for calls. Specifically, $G_i^k = \{j \in \{1, \dots, J\} : \gamma_{jk} = i\}$. S_i can also be expressed as the product of the utilization of unit i at calls and the service rate of unit i at calls, i.e.,

$$S_i = \rho_i \mu_i. \quad (19)$$

Larson [8] developed an approximation for p_{ijk} by assuming independence of servers and applying a correction factor.

$$p_{ijk} \approx Q(N, \bar{\rho}, k-1) \prod_{l=1}^{k-1} \rho_{\gamma_{jl}} (1 - \rho_i), \quad (20)$$

where $Q(N, \bar{\rho}, \cdot)$ is a correction factor for server dependence and $\bar{\rho}$ is the average utilization of all units. We have

$$\bar{\rho} = \frac{\lambda}{N\mu} (1 - P_b), \quad (21)$$

where P_b is the blocking probability that all units that can serve a call are busy. Define $P(k)$ as the probability that exactly k servers are busy at calls. The correction factor is given by

$$Q(N, \bar{\rho}, r) = \sum_{k=r}^{N-1} \frac{\binom{k}{r}}{\binom{N}{r}} \frac{N-k}{N-r} \frac{P(k)}{\bar{\rho}^r (1 - \bar{\rho})}. \quad (22)$$

The probabilities $P(k)$ and $P_b = P(N)$ are obtained by the Erlang loss formula.

Now, we have all the components needed to estimate ρ_i . Using (18), (19), and (20), we have

$$\rho_i = \frac{S_i}{\mu_i} = \frac{1}{\mu_i} \sum_{k=1}^N \sum_{j \in G_i^k} \lambda_j p_{ijk} = \frac{1}{\mu_i} \sum_{k=1}^N \sum_{j \in G_i^k} \lambda_j Q(N, \bar{\rho}, k-1) \prod_{l=1}^{k-1} \rho_{\gamma_{jl}} (1 - \rho_i), \quad (23)$$

where we obtain $Q(N, \bar{\rho}, k)$ from (22). Dividing both sides of (23) by $1 - \rho_i$ and collecting terms of ρ_i , the unit utilizations for calls are approximated by the following N non-linear equations,

$$\rho_i = 1 - \left(1 + \frac{1}{\mu_i} \sum_{k=1}^N \sum_{j \in G_i^{1,k}} \lambda_j Q(N, \bar{\rho}, k) \prod_{l=1}^{k-1} \rho_{\gamma_{jl}} \right)^{-1}, \quad i = 1, 2, \dots, N \quad (24)$$

These N non-linear equations can be solved iteratively to obtain ρ_i . In contrast, the exact solution requires solving a detailed balance equation consisting of 2^N equations. Recall that the objective function is given by $\min_{\mathbf{x}} \sum_{i \in I} \sum_{j \in J} t_{ij} q_{ij}(\mathbf{x})$. Denote the estimated unit utilization as $\hat{\rho}$. The hypercube approximation algorithm provides an estimate of q_{ij} . We have

$$\hat{q}_{ij} = \begin{cases} \frac{\lambda_j}{\sum_j \lambda_j} (1 - \hat{\rho}_i), & i = \gamma_{j1}, \\ \frac{\lambda_j}{\sum_j \lambda_j} Q(N, \bar{\rho}, k-1) \left(\prod_{l=1}^{k-1} \hat{\rho}_{\gamma_{jl}} \right) (1 - \hat{\rho}_i), & i = \gamma_{jk}, \forall k \geq 2. \end{cases} \quad (25)$$

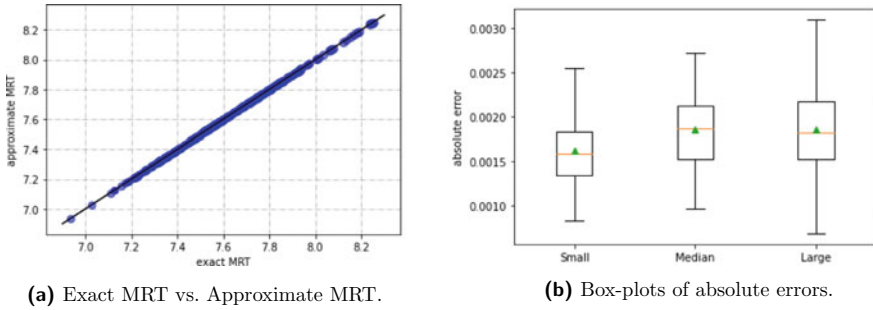


Fig. 32.1 The accuracy of the spatial hypercube approximation algorithm

We show that the approximate MRT is very close to the exact value in Fig. 32.1, with an mean absolute error of less than 0.002 min, by generating 100 random setups under small, median, and large problems. Such a small difference is assumed to be negligible in solving for the optimal solution.

Numerical Results

In this section, we perform numerical experiments to compare the performance of p-median, our Bayesian optimization approach, and the optimal solution via full enumeration. We constructed a hypothetical region, divided into a 9×9 grid, and randomly place the N candidate locations within the region. The results are displayed in Fig. 32.2.

We show that our algorithm always outperforms the p-median solution and converges very fast in numerical experiments. The average performance of the algorithm in each round and the 95% confidence interval are plotted in the figure. In small-sized problems (Fig. 32.2a, b), we can obtain the global minimum MRT by enumeration. The results show that our method can converge to the optimal solution in very few iterations. In the medium and large problems (Fig. 32.2c, d), we cannot obtain optimal solutions by enumeration, but our method can still surpass the p-median solution quickly. In our large of selecting 50 unit locations from 100 candidate locations, obtaining the optimal solution requires enumerating all 1.01×10^{29} possible solutions. In comparison, our method outperforms the p-median solution in about 90 iterations and converges in about 300 iterations.

Another noteworthy phenomenon is that there is a cusp point at around 20 iterations in all experiments, which means that we always observe a significant drop in MRT after 20 iterations. This is where the algorithm completes the initial exploration sampling phase and enters the Bayesian optimization phase, which means that our algorithm can learn some structure of the problem during the exploration phase, and the following suggested point leverages this learned information.

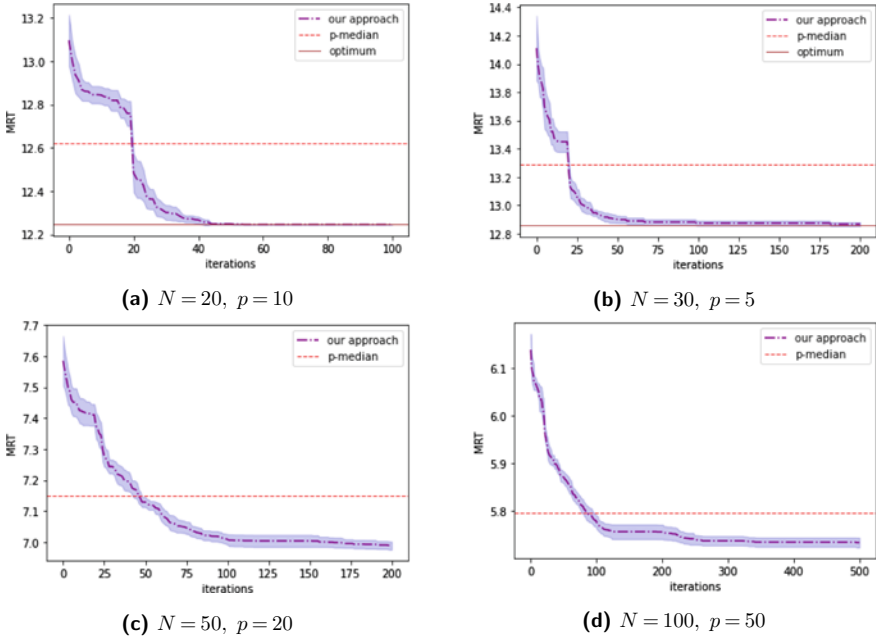


Fig. 32.2 Numerical experiments under different settings

Case Study: St Paul, Minnesota

We collected data on the emergency medical services in St. Paul, Minnesota. According to the most recent census, the population of St. Paul was 285,068 in 2017. We used 2014 Fire Department data for our analysis. In that year, 22,536 medical incidents were reported. This dataset includes the corresponding arrival times of each call, location of the nearest street intersections, and dispatch determinant categories that reveal the priority level of each call. We did not have personally identifiable information about each incident. We divided the map of St. Paul into 71 regions according to census tracts, as displayed in Fig. 32.4. We used Google Maps API to determine the travel time from each tract center to each candidate location, considering one-way streets and other practical issues. In St. Paul, there are 17 existing locations and 11 ambulances. Our goal is to use real data from St. Paul to place these ambulances in candidate locations to reduce the average response time to medical events across the city.

In Fig. 32.3, we plot the results and the best solution for ten runs of our method. Our method converge to the optimal value in the first 50 iterations. The optimal solutions obtained from our method and the p-median are shown in Fig. 32.4. We find that both our method and the p-median give optimal results. This is because St. Paul is a rather small city and the system has relatively little traffic compared to other large cities or cities with limited resources. Our solution can be applied even to the largest cities or countries.

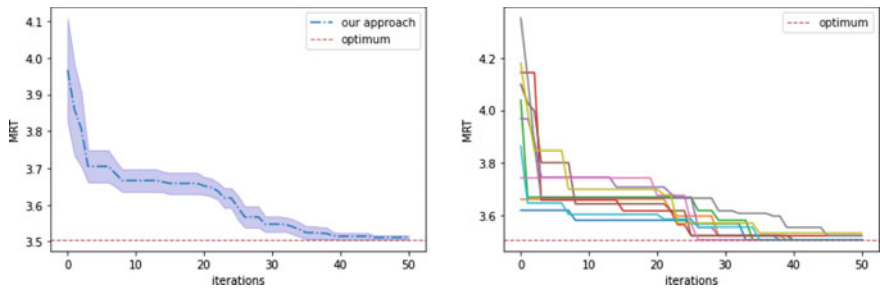


Fig. 32.3 Our solution approach applied to St. Paul, Minnesota

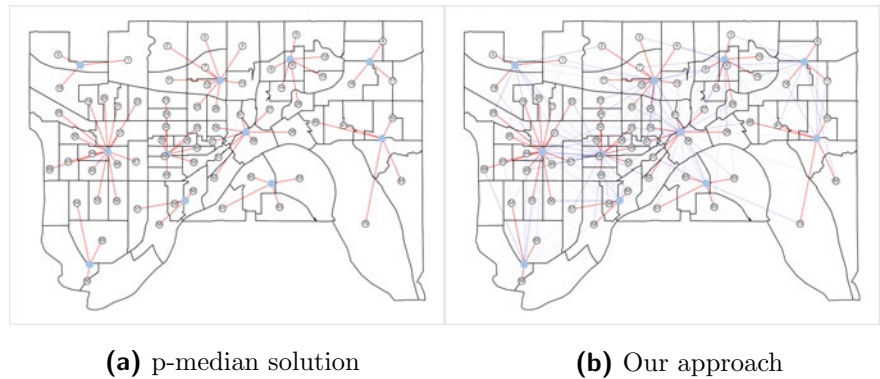


Fig. 32.4 Our approach versus p-median in St. Paul, Minnesota case study. The solid circles indicate unit locations, and the darknesses of the lines indicate the percentage of calls responded by each unit in each subregion. In p-median, there is only one line connected to each subregion, while in p-MRT, there are multiple lines per subregion, indicating that a call may be responded by other units when the closet unit is busy. The fraction of calls in each subregion responded to by the primary unit is marked in red, while the rest are marked in blue

Conclusion

In this paper, we propose a Bayesian optimization scheme to solve the optimal unit locations in emergency service systems that minimizes the system-wide mean response time. We develop both lower and upper bounds from the classical p-median problem. We use feasible trust region and adaptive swapping in searching for a solution and use a spatial hypercube approximation algorithm to obtain an estimate of the objective function. We show that our algorithm always converges to the global optimal solution, with a regret bound guarantee. In numerical experiments, we show that our algorithm always outperforms the p-median solution and converges very fast to the optimal solution. Our approach outperforms the p-median solution in less than 90 evaluations, and converges and yields a better than the p-median solution in less

than 300 evaluations. We also apply our approach in solving the best unit locations in St. Paul, Minnesota, using one-year real data. We show that our approach rapidly converges to the optimal solution.

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